

# **Customer & Sales Performance Analysis**

**by**

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## **1.Executive Summary**

This project analyzes retail sales data to identify customer purchasing patterns, revenue trends, and product performance. Python, SQL Server, and Microsoft Power BI were used for data cleaning, analysis, visualization, and dashboard creation. The findings provide insights that can help business improve sales performance and make data-driven decisions.

## **2.Business Problem**

The business wants to understanding customer purchasing behavior, identify the highest-performing products categories, analyze monthly revenue trends, and determine which customer groups contribute the most revenue. These insights will help improve marketing strategies and business decisions.

## **3.Data Cleaning**

Before analysis, the dataset was inspected for missing values, duplicate rows, suspicious values, confirmation of data types, verification of the date time column and inconsistencies between column. After the inspection, the cleaned data was exported SQL analysis.

## **4.Analysis**

The following analysis were carried out in SQL:

- Total Revenue

- Average Purchase Amount
- Total Number of Transactions
- Revenue by Product Category
- Revenue by Gender
- Top 10 Customers by Spending
- Monthly Revenue Trend
- Revenue by Age Group
- Highest Spending Product Category
- Average Customer Age

## 5. Insights

- Revenue peaked in May and October.
- Electronics generated the highest revenue.
- Male and female customers spent similar amounts.
- A small number of customers contributed to a large share of total sales.
- Customers aged 18-25 contributed to the highest revenue.

## 6. Recommendations

- Launch promotions during low sales month.
- Increase inventory for electronics due to stronger sales.
- Develop loyalty programs for high-value customers.
- Target the highest-spending age group with personalized marketing.
- Continue monitoring monthly revenue to identify seasonal trends.

## 7. Conclusion

This analysis provided valuable insights into customer purchasing behavior and overall sales performance. Using Python, SQL, and Power BI, key business metrics were identified and presented through interactive dashboards. The findings can support better business decisions and demonstrate the value of data-driven analysis.